KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2019
Department of Computer Science and Engineering
CSE 4111
Machine Learning

TIME: 3 hours

FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

## SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) What do you mean by learning? In biological system, what are the elements and how they response for learning? (08)
   b) Draw computational model of a single neuron. Solve different binary logics with the model. (09)
   c) What are the different types of memory? What are their significances and uses? (09)
   d) Calculate the weight set (W) of a Hetero-Associative memory for the following samples: (09)

| Pattern, p | Source s(p) | Target, t(p) |
|---|---|---|
| 1 | 1 0 0 0 | 1, 0 |
| 2 | 1 1 0 0 | 1, 0 |
| 3 | 0 0 1 1 | 0, 1 |
| 4 | 0 0 0 1 | 0, 1 |

   Finally verify the system with appropriate samples.

2. a) What do you understand by linear problem and non-linear problem in the context of machine learning? Why solving non-linear problem is challenging? Briefly explain with an appropriate example. (07)
   b) "An additional dimension (input) may transform a non-linear problem to linearly separable"- justify the statement with proper example. (08)
   c) Why training may not perform after certain iteration even though error remain in training a neural network using Back-Propagation? Explain with appropriate example. (10)
   d) What is the condition to use a function as Activation Function in neural network? Graphically represent two different forms of sigmoidal functions and their derivatives. Explain similarity and dissimilarity between the functions. (10)

3. a) What are the basic differences between Radial Basis Function (RBF) network and Multi Layer Perceptron (MLP)? Briefly describe training steps of RBF networks. (09)
   b) Why ensemble of neural networks is essential when an MLP solve any arbitrary function? What are the conditions for ensemble construction? (06)
   c) What are the different techniques for training individual networks in ensemble construction? Mention different ensemble methods for each type. (08)
   d) Describe an ensemble method which uses sampling of individual patterns. Is such technique suitable for ensemble of Decision Tree? Explain briefly. (12)

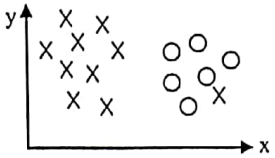
4. a) What do you mean by Swarm Intelligence (SI)? Briefly describe common features of SI based algorithms. (05)
   b) Explain importance of exploration and exploitation properties in SI algorithms. Describe exploration and exploitation matter in Particle Swarm Optimization (PSO). (10)
   c) In Ant Colony Optimization (ACO), how an ant complete a tour in solving Travelling Salesman Problem (TSP)? (08)
   d) What do you understand by the terms Global Search, Local Search and Blind Search? Does GA follows the search methods? Explain your justification with techniques included in GA. (12)

## SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) What are support vectors in Support Vector Machine (SVM)? How does Linear SVM work? Explain. (10)
   b) How can SVM classify non-linearly separable data? Explain with an example and proper figure. (07)
   c) Can SVM be over trained? What can be the effect of over training? How will you detect over fitting of a SVM model? (08)

d) Can you classify the data points (x, 0) depicted in the following figure using LSVM?




Explain the procedure of finding the correct hyper plane for the data points.

6. a) Define "Occam's Razor Principle". Explain its effect on Decision Tree. (07)
   b) "Entropy measures the impurity of training data instances"-is this statement true or false? Explain with proper reasoning. (08)
   c) Formulate a decision tree based on the following experience table where Buys-computer is set as target class. (20)

| Records | Age | Income | Student | Credit-Rating | Buys-computer |
|---|---|---|---|---|---|
| 1 | <= 30 | High | No | Fair | No |
| 2 | <= 30 | High | No | Excellent | No |
| 3 | 31 … 40 | High | No | Fair | Yes |
| 4 | > 40 | Medium | No | Fair | Yes |
| 5 | > 40 | Low | Yes | Fair | Yes |
| 6 | > 40 | Low | Yes | Excellent | No |
| 7 | 31 … 40 | Low | Yes | Excellent | Yes |
| 8 | <= 30 | Medium | No | Fair | No |
| 9 | = 30 | Low | Yes | Fair | Yes |
| 10 | > 30 | Medium | Yes | Fair | Yes |
| 11 | <= 30 | Medium | Yes | Excellent | Yes |
| 12 | 31 … 40 | Medium | No | Excellent | Yes |
| 13 | 31 … 40 | High | Yes | Fair | Yes |
| 14 | > 40 | Medium | No | Excellent | No |

7. a) Why Pre-pruning and Post-pruning are necessary for decision trees? How does error propagate in decision tress? (10)
   b) How do you tackle the curse of dimensionality? (05)
   c) Write short notes on: (i) Data condensation, and (ii) Data squashing. (08)
   d) How does Principle Component Analysis construct the principle components? Point out the steps briefly with an example. (12)

8. a) Consider the data in the following table: (15)

| ID | Height | Age | Weight |
|---|---|---|---|
| 1 | 5 | 45 | 77 |
| 2 | 5.11 | 26 | 47 |
| 3 | 5.6 | 30 | 55 |
| 4 | 5.9 | 34 | 59 |
| 5 | 4.8 | 40 | 72 |
| 6 | 5.8 | 36 | 60 |
| 7 | 5.3 | 19 | 40 |
| 8 | 5.8 | 28 | 60 |
| 9 | 5.5 | 23 | 45 |
| 10 | 5.6 | 32 | 58 |
| 11 | 5.5 | 38 | ?? |

Predict the weight of ID 11 based on their height and age using KNN.

   b) What are the strengths of K-means algorithm? (05)
   c) Differentiate between divisive clustering and agglomerative clustering. (08)
   d) Draw a dendrogram from the following figure: (07)

